MicroArray-based Methylation2Activity: Advancing biological knowledges from DNA methylation profiles of patient tumors

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BACKGROUND

DNA methylation (DNAm) is a well-studied and relatively stable regulatory epigenetic mechanism that can be accurately and robustly profiled in various tissues. Multiple studies have shown that DNAm predicts chromatin packaging patterns and reveals prognostically significant and distinct cancer subgroups. However, its transcriptional regulatory roles is neither linear nor simple, which has severely hampered the understanding of DNAm signature's biological significance. To address this challenge, we have developed and published a deep-learning framework MethylationToActivity (M2A) that accurately predicts individual promoter's activity (measured as H3K27ac enrichment) from WGBS methylomes. On the other side, array-based methylome is more cost effective, making it the most common epigenetic data collected from patients' tumor samples despite a limited coverage of CpGs. Therefore, in this study, we redesigned M2A to infer promoter enrichment from array methylomes.

METHODS

Number of windows tested:
10, 20, 40, 80*, 160*, 320*, and 640*
[* = additional Max Pooling to layers].

Kernel sizes tested:
3, 5, 7, 9, 11, and 13

Features tested:
3 and 4 (described further in Results)

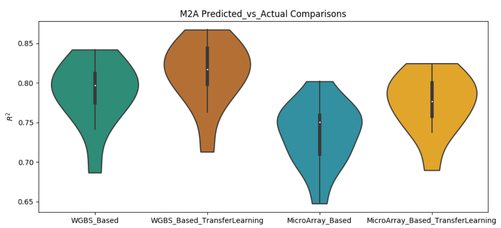
Window resolutions tested:
[250bp, 2500bp] and [500bp,5000bp]

Learning rates tested:
0.005, 0.01, and 0.02

Table 1: M2A Parameter Tuning.
R^2 = Actual-vs-Predicted Pearson Correlation; *=Parameter being tested.

	Current Model	Run Details				
Number of Windows	320	320	160*	640*	320	320
Number of Features	4	3*	4	4	4	4
Kernel Size	[11,1]	[11,1]	[11,1]	[11,1]	[9,1]*	[13,1]*
Sample	R^2					
NBL_11_2_FFPE	0.7589	0.757	0.7568	0.7557	0.7577	0.7572
NBL_11_5_Frozen	0.7851	0.7844	0.7838	0.7819	0.7867	0.7858
NBL_268_FFPE	0.8471	0.8417	0.847	0.8421	0.8448	0.8478
NBL_3_1_Frozen	0.8922	0.8884	0.8892	0.8882	0.8886	0.8932
NBL_78_b23_FFPE	0.9079	0.8996	0.9047	0.9010	0.9011	0.9049
Average	**0.8382**	**0.8342**	**0.8363**	**0.8338**	**0.8358**	**0.8378**
Standard Deviation	**0.0582**	**0.0561**	**0.0577**	**0.0572**	**0.0559**	**0.0581**

ACCURACY



	WGBS	WGBS_TL	MicroArray	MicroArray_TL
count	13	13	13	13
mean	0.7878	0.8116	0.7353	0.7748
std	0.0417	0.0413	0.0400	0.0374
min	0.6866	0.7129	0.6476	0.6897
25%	0.7747	0.7982	0.7097	0.758
50%	0.7971	0.8171	0.7504	0.7768
75%	0.8122	0.8439	0.7593	0.7999
max	0.8421	0.8673	0.802	0.8247

Figure 1 & Table 2: M2A Summary Statistics (WGBS vs MicroArray).
Actual-vs-Predicted Pearson Correlation Violin Plot and corresponding Summary Statistics Table; WGBS/WGBS_Based = WGBS-based M2A vanilla model; WGBS_TL/ WGBS_Based_TransferLearning= WGBS-based M2A model with transfer learning; MicroArray/MicroArray_Based = MicroArray-based M2A vanilla model; MicroArray_TL/ MicroArray_Based_TransferLearning= MicroArray-based M2A model with transfer learning; R^2 = Actual-vs-Predicted Pearson Correlation

MODEL OVERVIEW & COMPARISON

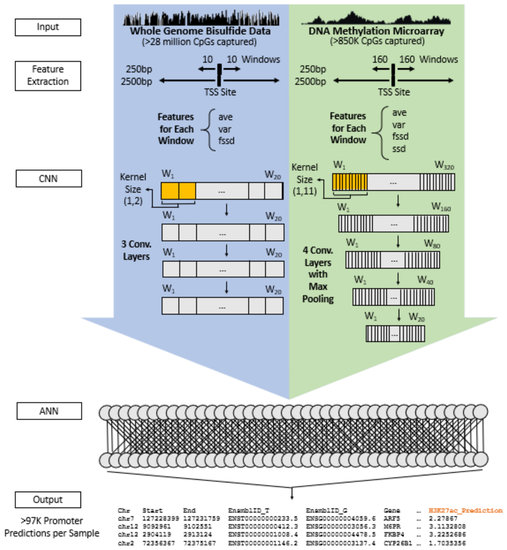


Figure 2: Side-By-Side Visual Comparison of WGBS-based M2A vs MicroArray-based M2A.

BIOLOGICAL SIGNIFICANCE

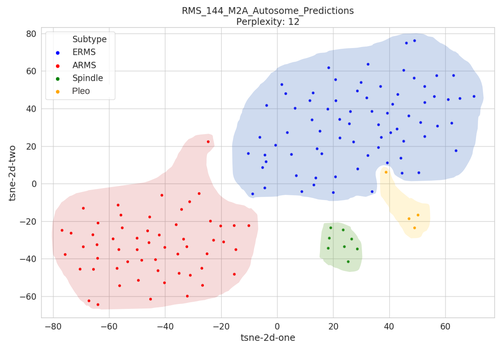


Figure 3: Rhabdomyosarcoma MicroArray Cohort M2A predictions tSNE.
Subtypes did cluster based on autosomal promoter enrichment prediction; N=144; perplexity =12;
ERMS = embryonal RMS;
ARMS = alveolar RMS;
Spindle = spindle cell/sclerosing RMS;
Pleo = pleomorphic RMS

BIOLOGICAL SIGNIFICANCE (cont.)

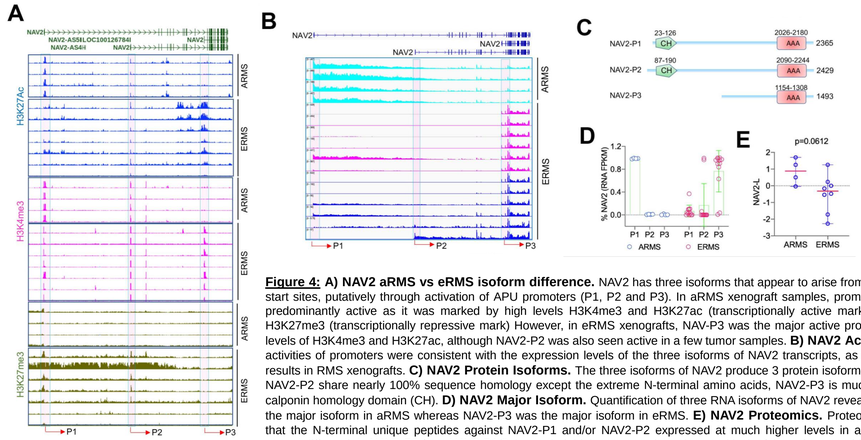
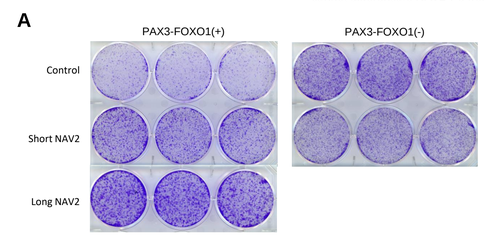
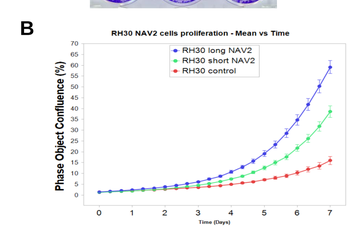


Figure 4: A) NAV2 aRMS vs eRMS isoform difference. NAV2 has three isoforms that appear to arise from different transcription start sites, putatively through activation of APU promoters (P1, P2 and P3). In aRMS xenograft samples, promoter 1 (NAV2-P1) was predominantly active as it was marked by high levels H3K4me3 and H3K27ac (transcriptionally active marks), and low levels of H3K27me3 (transcriptionally repressive mark) However, in eRMS xenografts, NAV-P3 was the major active promoter defined by high levels of H3K4me3 and H3K27ac, although NAV2-P2 was also seen active in a few tumor samples. **B) NAV2 Active Promoters.** The activities of promoters were consistent with the expression levels of the three isoforms of NAV2 transcripts, as indicated by RNA-seq results in RMS xenografts. **C) NAV2 Protein Isoforms.** The three isoforms of NAV2 produce 3 protein isoforms. While NAV2-P1 and NAV2-P2 share nearly 100% sequence homology except the extreme N-terminal amino acids, NAV2-P3 is much shorter and lacks a calponin homology domain (CH). **D) NAV2 Major Isoform.** Quantification of three RNA isoforms of NAV2 revealed that NAV2-P1 was the major isoform in aRMS whereas NAV2-P3 was the major isoform in eRMS. **E) NAV2 Proteomics.** Proteomics analysis showed that the N-terminal unique peptides against NAV2-P1 and/or NAV2-P2 expressed at much higher levels in aRMS than eRMS (we cannot differentiate NAV2-P3 from NAV2-P1/2 due to their high sequence homology and lack of the unique peptides against NAV2-P3).





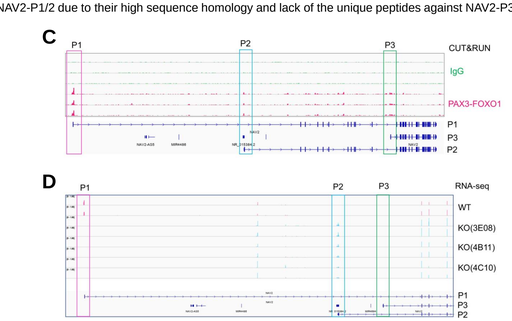


Figure 5: NAV2 overexpression increases cell proliferation in PAX3-FOXO1 positive cells. A) Cell proliferation. RH30 (P3F1 positive) cells show higher proliferation with NAV2 overexpression, whereas RD cells (P3F1 negative) did not show any effect. **B) Proliferation Over Time.** IncuCyte experiment showed RH30 cells proliferate more than control cells. **PAX3-FOXO1 regulates the expression of NAV2-P1 isoform. C) PAX3-FOXO1 binding peaks.** PAX3-FOXO1 exhibited a strong binding peak at NAV2-P1, but not at NAV2-P2 and NAV2-P3 regions. **D) PAX3-FOXO1 KO.** RNA-seq analysis of Rh30 cells after CRISPR-Cas9 mediated PAX3-FOXO1 knockout revealed that the expression of the first exon in NAV2-P1 disappeared. Intriguingly, the first exon of NAV2-P2 had a marked induction, indicating that there is an APU event triggered by loss of PAX3-FOXO1, leading to the switch from NAV2-P1 to NAV2-P2.

BIOLOGICAL EXAMPLE CONCLUSION

mRNA and protein diversity can be enormously increased by alternative promoter usage (APU), and these mechanisms could be "hijacked" by cancer cells for promoting tumorigenesis. M2A analysis comparing eRMS and aRMS RMS subtypes revealed two major isoforms of NAV2 existing in aRMS and eRMS due to APU. Molecular experiments revealed that PAX3-FOXO1 directly binds to the aRMS-specific promoter and is required for its active transcription. Genetic deletion of PAX3-FOXO1 leads to NAV2 APU. Overexpression of aRMS-specific NAV2 isoform significantly promoted cell proliferation, supporting its oncogenic function. This work reveals that APU is one mechanism that contributes to novel cancer related genes in RMS, which provides a new insight into understanding RMS pathogenesis and additional therapeutic strategies.

MAIN CONCLUSION

Micro-Array Based M2A was effective in predicting promoter enrichment with an accuracy comparable to its predecessor, the whole-genome bisulfide-based M2A model, even with 97% less data input from the methylation platform. This successful model now has the power to profoundly influence the study of epigenetic deregulation by allowing for promoter enrichment to be derived from the most commonly collected epigenetic data, DNAm, providing vast insight into downstream promoter consequences.

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